



FACULTÉ DES SCIENCES ET DES TECHNOLOGIES(FST)

TD N°6 – Réseaux

COURS: Réseaux-2

PRÉPARÉ PAR: Brian ANTOINE

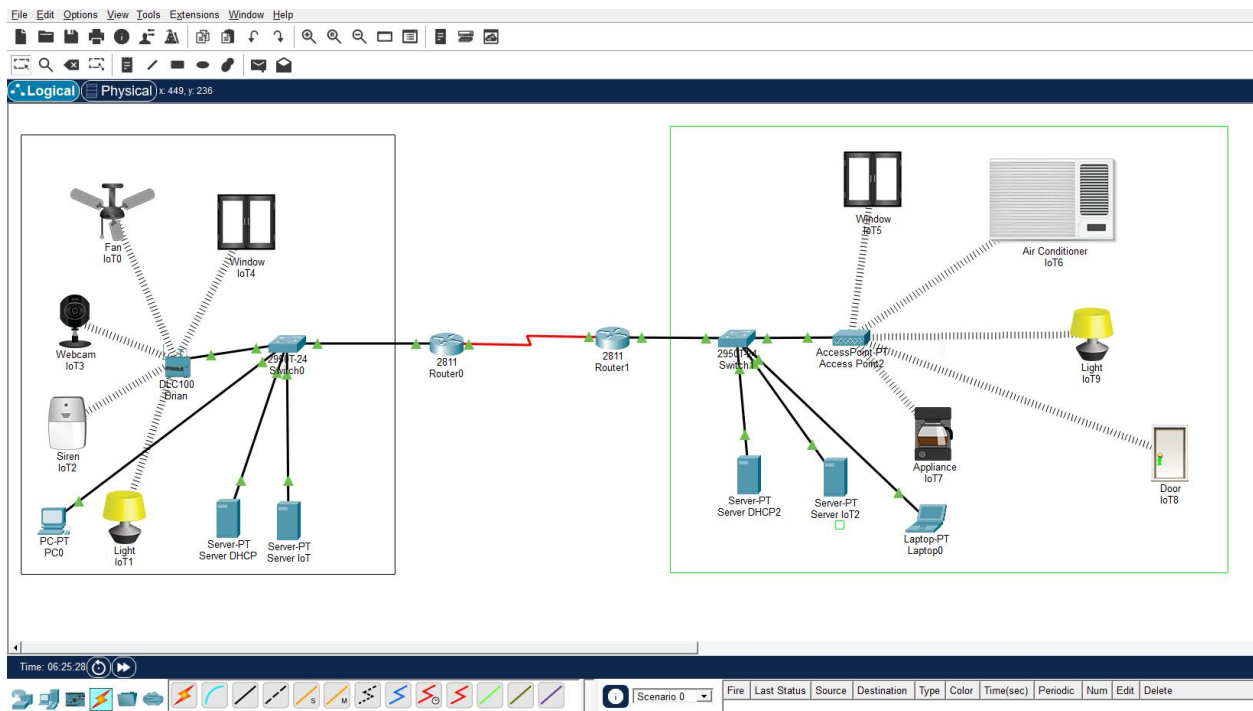
NIVEAU: L3

DATE: 03/06/2026

Objectifs

1. Le contrôle à distance
2. La communication entre objets
3. La sécurité du réseau IoT

Configuration d'un Réseau Smart Home (IoT)



Configuration de R1

Press RETURN to get started!

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 192.168.1.1 255.255.255.252
Router(config-if)#clock rate 64000
This command applies only to DCE interfaces
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config)#
```

Copy

Paste

☐ Top

Configuration de R2

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface Serial0/3/0
Router(config-if)#ip address 192.168.1.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#exit
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config)#
```

Copy

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Routage entre les deux routeurs (OSPF)

R1

```
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.2.0 0.0.0.255 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

☐ Top

R2

```
*****
Router(config)#router ospf 1
Router(config-router)#network 192.168.1.0 0.0.0.3 area 0
Router(config-router)#network 192.168.3.0 0.0.0.255 area 0
Router(config-router)#do wr
Building configuration...
[OK]
Router(config-router)#
```

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Configuration des servers DHCP pour attribuer des IP automatiquement

DHCP1

Server DHCP

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DHCP

InterfaceFastEthernet0ServiceOnOff

Pool NameserverPool

Default Gateway192.168.2.1

DNS Server0.0.0.0

Start IP Address :19216823

Subnet Mask:2552552550

Maximum Number of Users :51

TFTP Server:0.0.0.0

WLC Address:0.0.0.0

AddSaveRemove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.2.1	0.0.0.0	192.168.2.3	255.255.2...	51	0.0.0.0	0.0.0.0

DHCP2

Server DHCP2

PhysicalConfigServicesDesktopProgrammingAttributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

PRP

DHCP

Interface

FastEthernet0

Service

On

Off

Pool Name

serverPool

Default Gateway

192.168.3.1

DNS Server

0.0.0.0

Start IP Address :

192

168

3

3

Subnet Mask:

255

255

255

0

Maximum Number of Users :

51

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

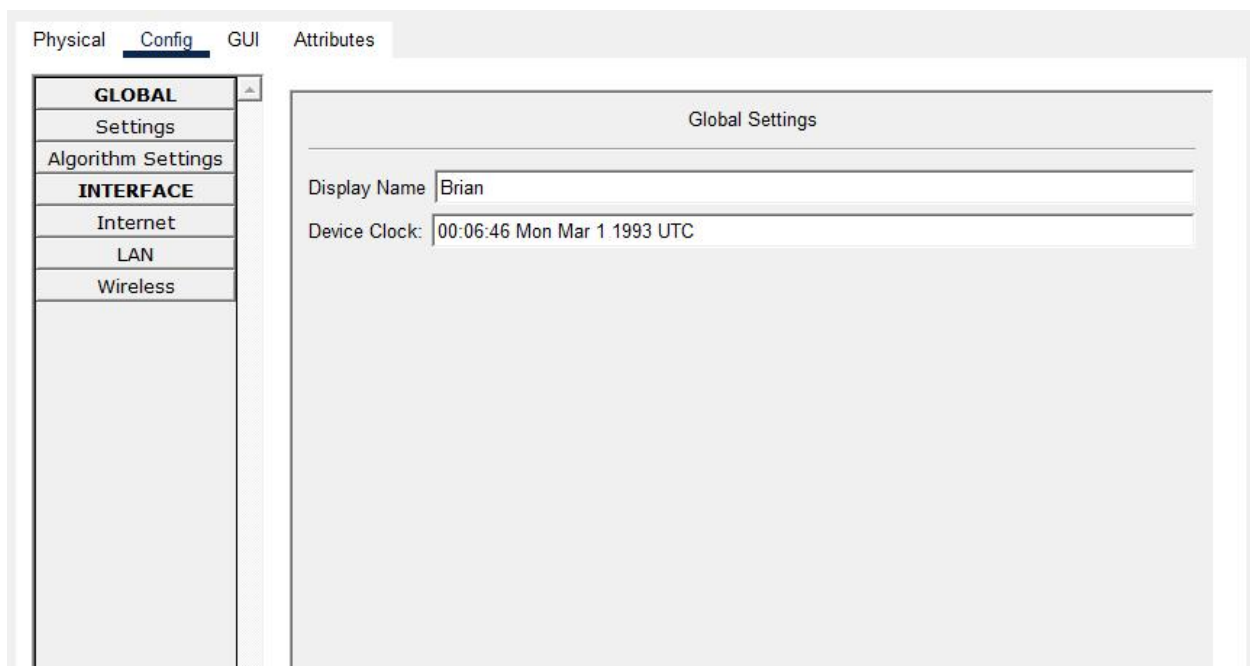
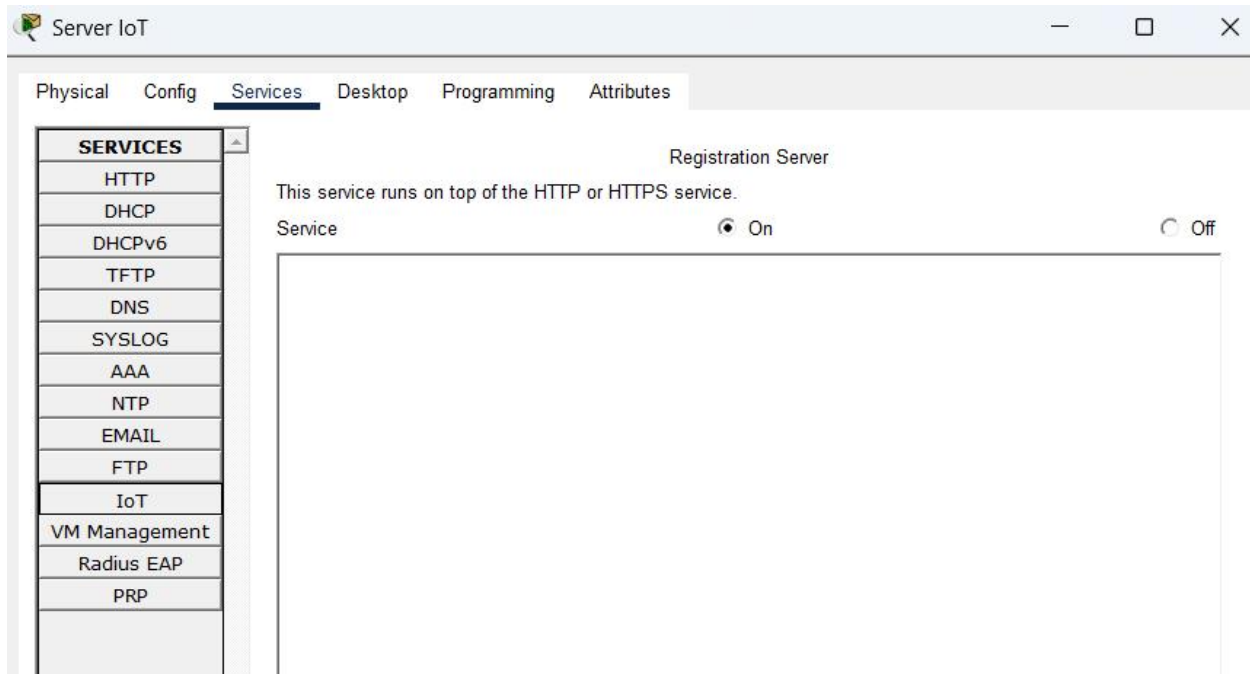
Save

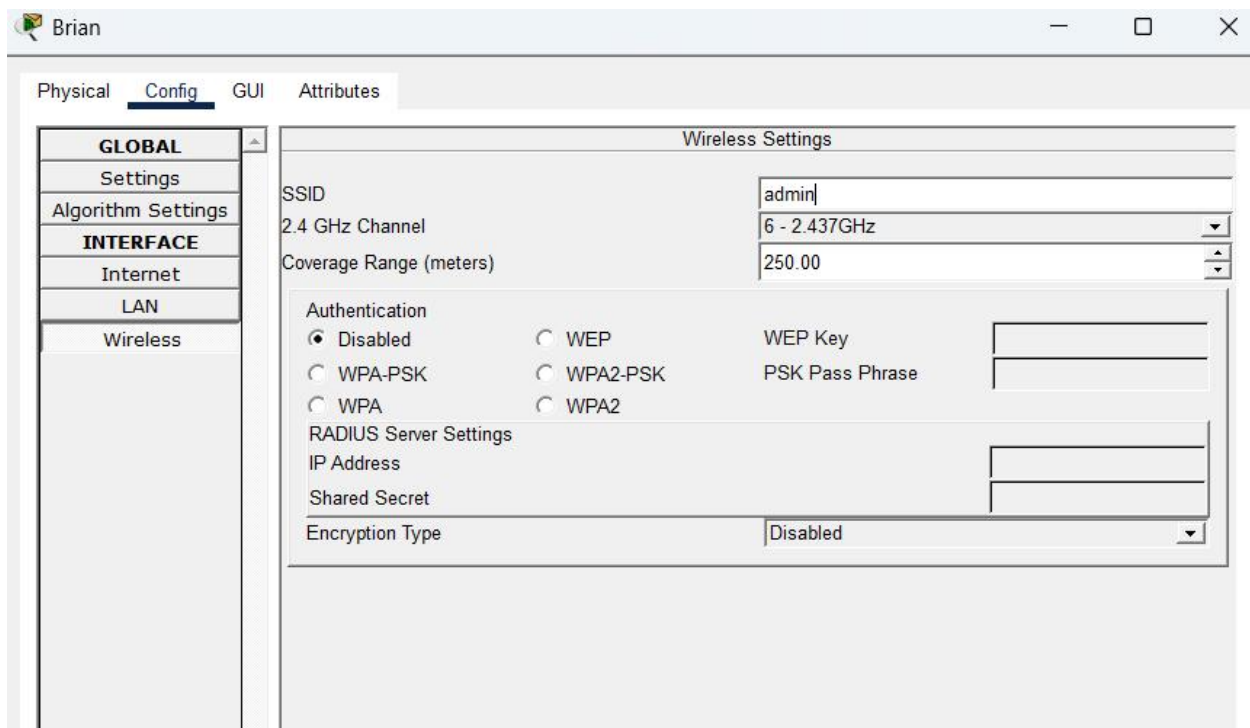
Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168.3.1	0.0.0.0	192.168.3.3	255.255.2...	51	0.0.0.0	0.0.0.0

Configuration du serveur IoT

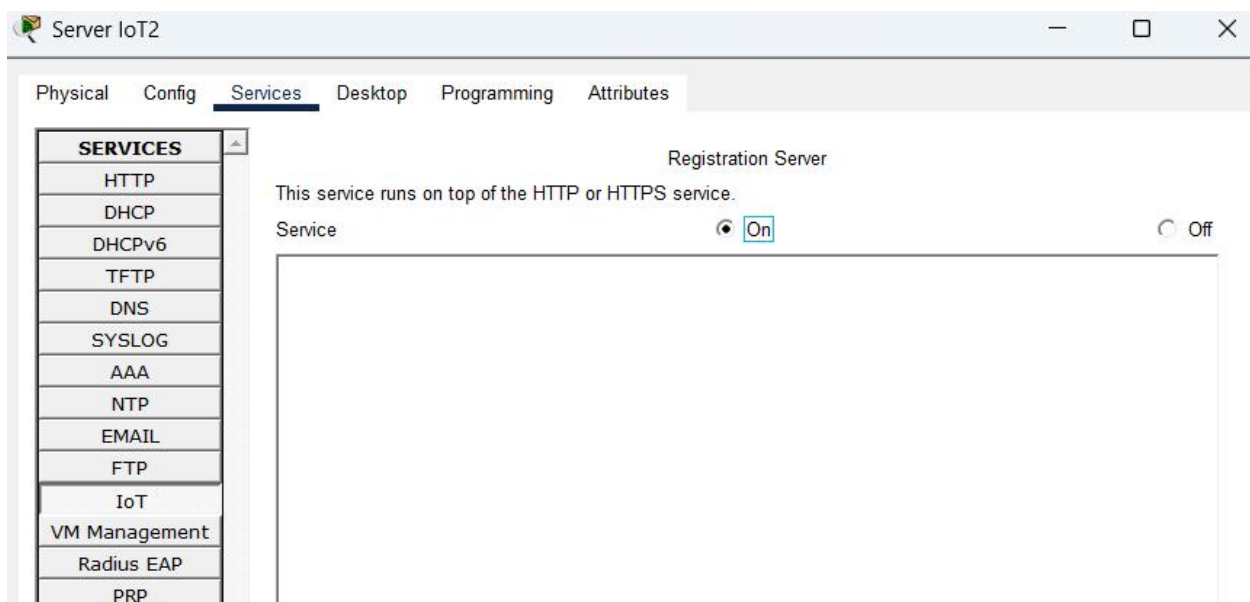
IoT1





1

IoT2



Access Point2

Physical **Config** Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port 1

Port Status ☒ On

SSID admin1

2.4 GHz Channel 6

Coverage Range (meters) 140.00

Authentication

☒ Disabled ☐ WEP ☐ WPA-PSK ☐ WPA2-PSK

WEP Key

PSK Pass Phrase

User ID

Password

Encryption Type Disabled

Registration au server

IoT1

Web Browser

< > URL http://192.168.2.3 Go Stop

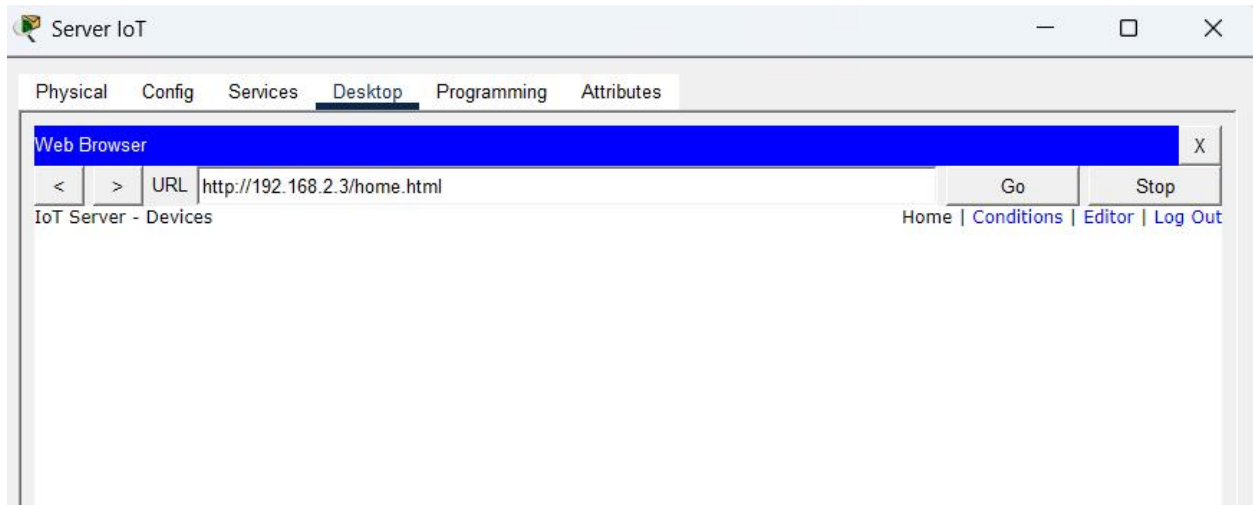
Registration Server Login

Username: admin

Password:

Sign In

Don't have an IoT account? [Sign up now](#)



Ajout des IoT

GLOBAL

Settings

Home Settings

Files

INTERFACE

Wireless0

Bluetooth

Global Settings

Display Name

Serial Number

Interfaces

Gateway/DNS IPv4

☒ DHCP

☐ Static

Default Gateway

DNS Server

Gateway/DNS IPv6

☒ Automatic

☐ Static

Default Gateway

DNS Server

IoT Server

☐ None

☐ Home Gateway

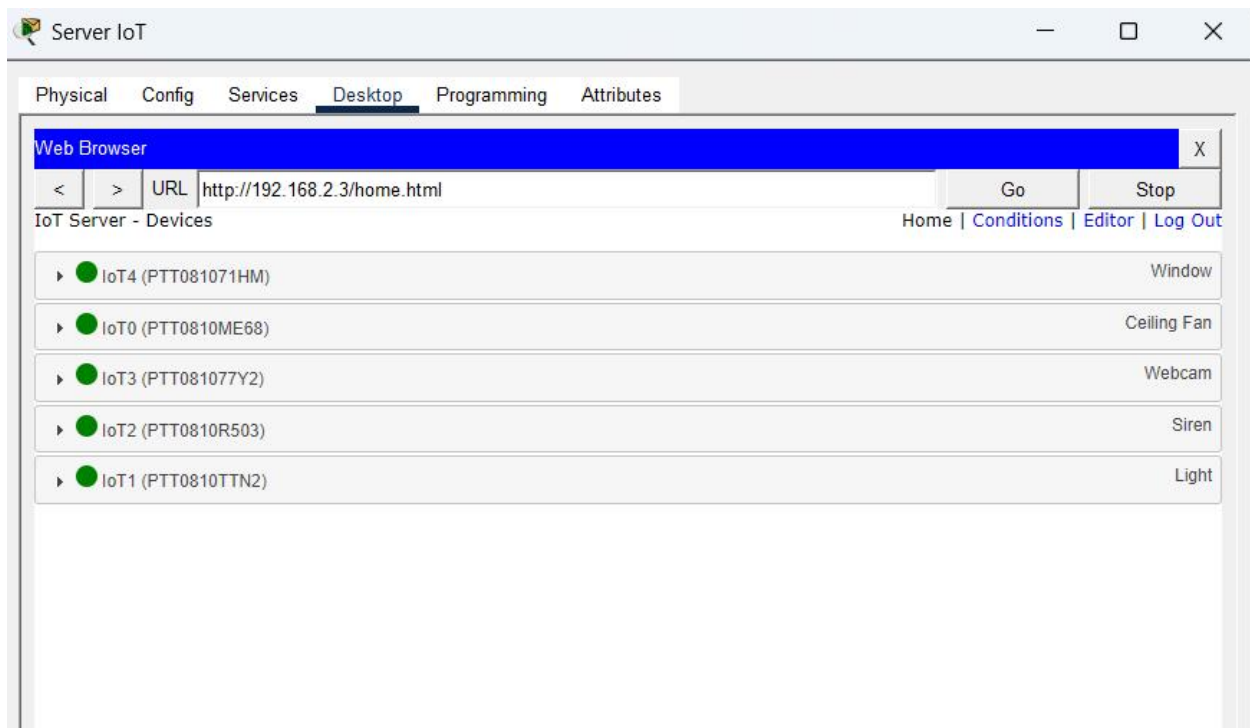
☒ Remote Server

Server Address

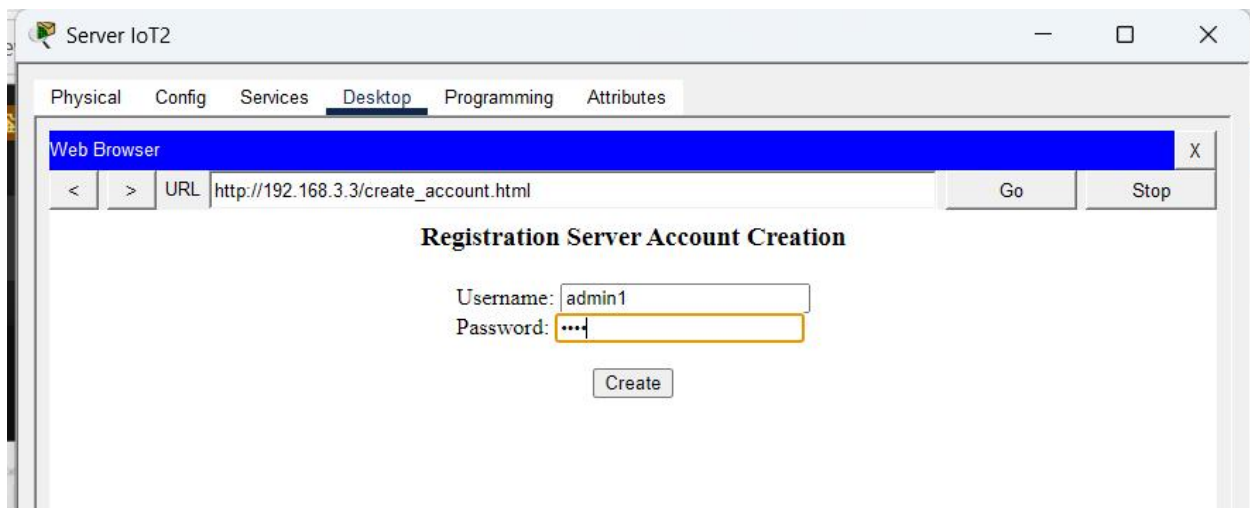
User Name

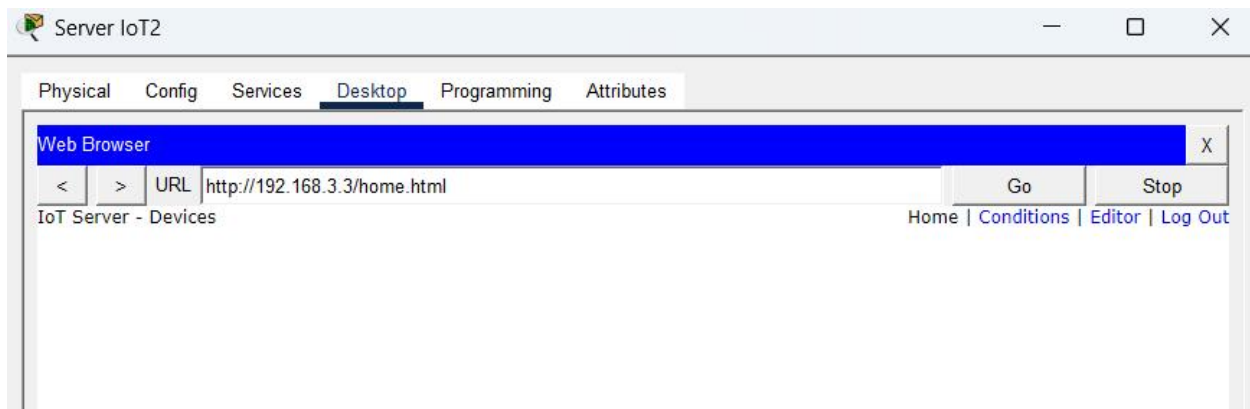
Password

Device Clock: 01:22:57 Mon Mar 1 1993 UTC



IoT2





Ajout des IoT

Specifications Physical **Config** Attributes

GLOBAL

Settings

Algorithm Settings

Files

INTERFACE

Wireless0

Bluetooth

Global Settings

Display Name

Serial Number

Interfaces

Gateway/DNS IPv4

☒ DHCP

☐ Static

Default Gateway

DNS Server

Gateway/DNS IPv6

☒ Automatic

☐ Static

Default Gateway

DNS Server

IoT Server

☐ None

☐ Home Gateway

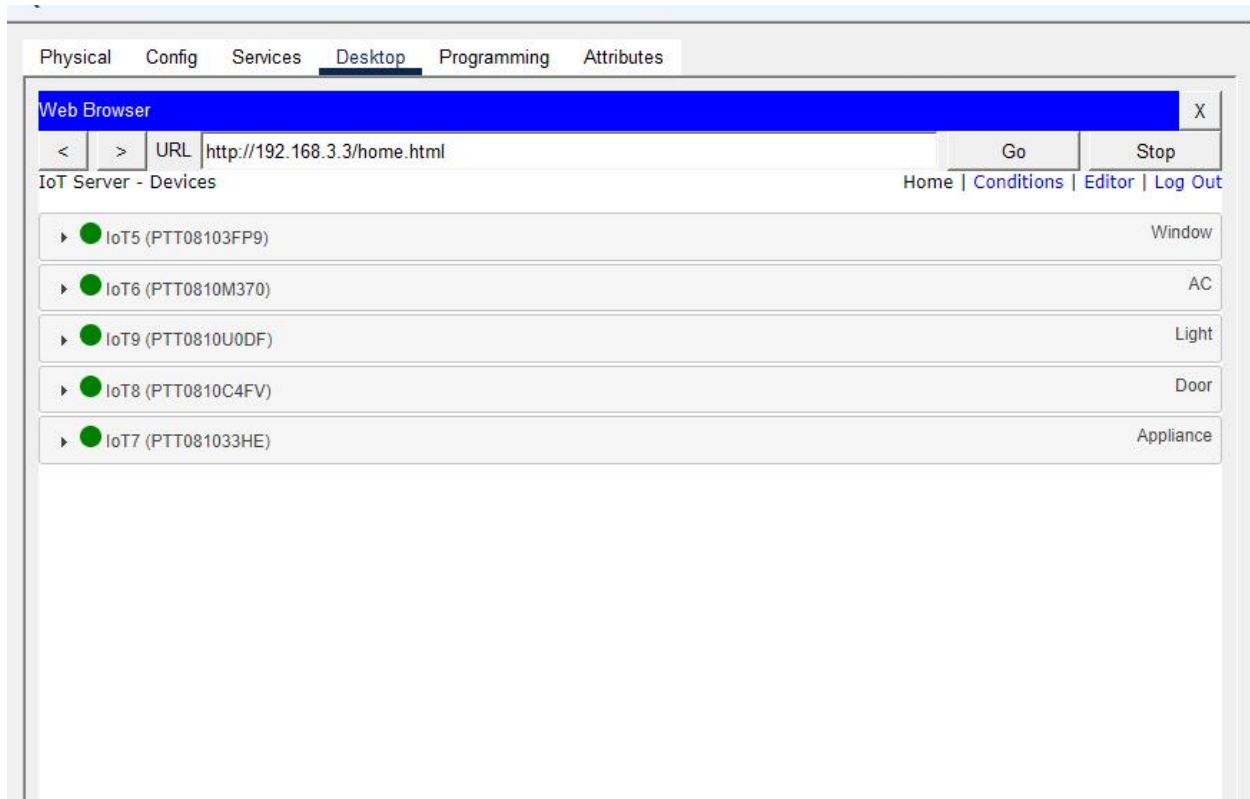
☒ Remote Server

Server Address

User Name

Password

Device Clock: 01:20:20 Mon Mar 4 1992 UTC



Automatisation (IoT Rules)

IoT1

Actions	Enabled	Name	Condition	Actions
---------	---------	------	-----------	---------

Add

Add Rule

Name

Enabled ☒

If:

Match

All

IoT4

On

 is

true

+ Condition

+ Group

Then set:

IoT0

Status

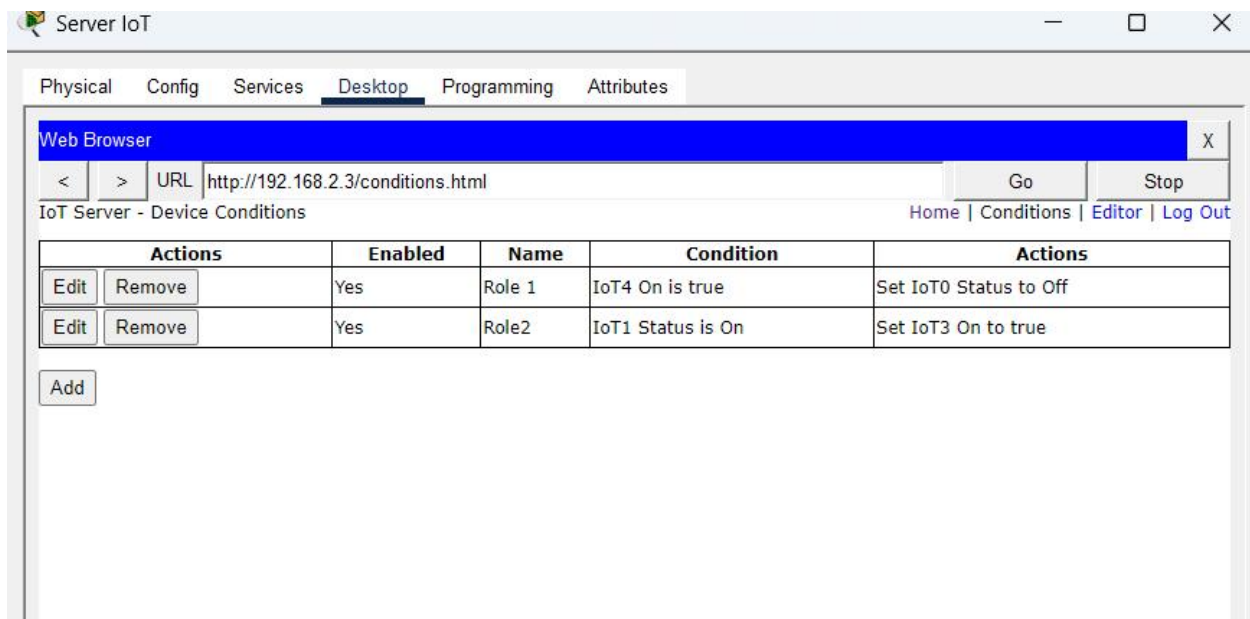
 to

Off

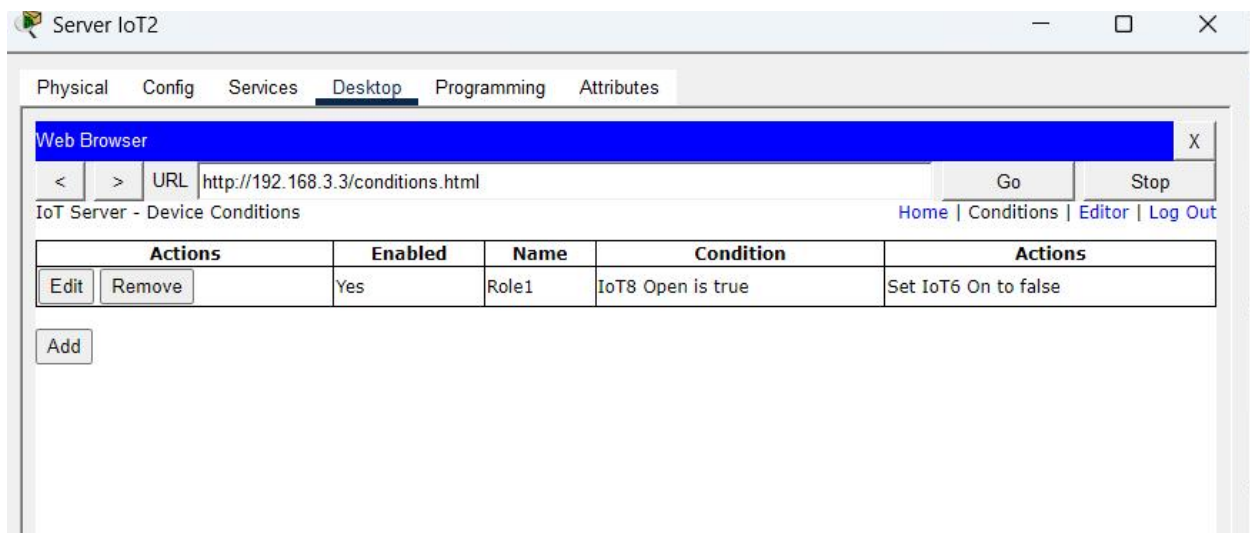
+ Action

OK

Cancel



IoT2



Conclusion

Grâce à ce TD, j'arrive à comprendre la communication entre objets, savoir configurer et sécuriser les réseau IoT.